

1. Administrative & Study Identification

Full Study Title: Tisagenlecleucel versus standard of care in adult patients with relapsed or refractory aggressive B-cell non-Hodgkin lymphoma: A randomized, open label, phase III trial **Short Title/Acronym:** BELINDA **Protocol Number:** CCTL019H2301 **NCT Number:** NCT03570892 **Trial Status:** Active, Not Recruiting **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Tisagenlecleucel (Kymriah, CTL019) | **ATC Code:** L01XL04 **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate and compare the efficacy of tisagenlecleucel with the standard of care (SOC) by measuring Event-Free Survival (EFS) in patients with relapsed or refractory aggressive B-cell Non-Hodgkin Lymphoma (NHL). **Secondary Objectives:** To evaluate Event Free Survival (EFS) as assessed by Local Investigator, Overall Survival (OS), Overall Response Rate (ORR), Duration of Response (DOR), Time to definitive deterioration in patient-reported outcomes (SF-36v2, FACT-Lym), and assess safety and tolerability.

2.2 Background & Rationale Approximately two-thirds of patients with B-cell NHL are cured with frontline chemoimmunotherapy. However, a subset of patients who do not respond to initial therapy or who relapse within 365 days following frontline therapy have poor outcomes and an expected survival of less than 12 months. In this subset, standard of care (SOC) includes platinum-based salvage chemotherapy followed by autologous hematopoietic stem cell transplant (HSCT) for chemo-sensitive disease. Since more than half of these patients fail to achieve adequate responses to reach HSCT, there is a critical unmet need. This study evaluates whether tisagenlecleucel (a CD19-directed CAR T-cell therapy) administered as a second-line therapy can improve outcomes for participants eligible for autologous HSCT.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard of Care: salvage chemotherapy followed by autologous HSCT) **Blinding:** Open-label **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Target **N**: 331 targeted enrollment. (Note: Approximately 318 subjects were planned to be randomized; 322 subjects were analyzed in the Full Analysis Set: 162 in the tisagenlecleucel arm and 160 in the SOC arm). **Number of Sites**: 72 locations globally **Estimated Study Start**: May 2019 **Estimated Primary Completion**: February 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult participants with Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1. 2. Histologically confirmed aggressive B-cell NHL that is relapsed/progressed within 365 days from the last dose of anti-CD20 antibody and anthracycline-containing frontline chemoimmunotherapy, or primary refractory disease. 3. Patient is considered eligible for autologous HSCT as per local investigator assessment. 4. Disease that is active on PET scan (5-Deauville score of 4 or 5) and measurable on CT scan (Nodal lesions >15 mm; Extranodal lesions >10 mm). 5. Adequate organ function: eGFR ≥ 60 mL/min/1.73m² or Creatinine $\leq 1.5 \times$ ULN; ALT/AST $\leq 5 \times$ ULN, Total bilirubin $\leq 1.5 \times$ ULN (Gilbert syndrome $\leq 3.0 \times$ ULN); ANC > 1000 /mm³, ALC > 300 /mm³ or CD3⁺ T cells > 150 /mm³, platelets $\geq 50,000$ /mm³, and hemoglobin > 8.0 g/dl; adequate pulmonary function (FEV1 $\geq 50\%$ and/or DLCO $\geq 50\%$, O2 sat $> 90\%$). 6. Leukapheresis material of non-mobilized cells available for manufacturing.

4.2 Exclusion Criteria 1. Prior treatment with anti-CD19 therapy, T-cell therapy, any prior gene therapy product, or prior allogeneic HSCT. 2. Active central nervous system (CNS) involvement (unless effectively treated >4 weeks prior to randomization) or clinically significant active infection. 3. Severe cardiovascular conditions including LVEF $< 45\%$, NYHA class III or IV within 12 months, unstable angina/MI/CABG/stroke within 6 months, significant arrhythmias, resting QTcF ≥ 450 msec (male) or ≥ 460 msec (female), or TdP risk factors. 4. Active neurological autoimmune or inflammatory disorders (e.g., Guillain-Barré Syndrome, ALS) or clinically significant active cerebrovascular disorders (e.g., PRES). 5. Treatment with any systemic lymphoma-directed second line anticancer therapy prior to randomization (only steroids and local irradiation permitted).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Lymphodepleting chemotherapy followed by a single intravenous (IV) infusion of $0.6-6.0 \times 10^8$ CAR-T cells. **Route of Administration**: Intravenous (IV). **Duration of Treatment**: Start of bridging chemotherapy (if applicable) until the infusion of tisagenlecleucel (expected on average at approximately 6 weeks from randomization).

5.2 Concomitant & Prohibited Therapy Permitted: Bridging chemotherapy while awaiting CAR-T cell manufacturing. **Prohibited:** Concomitant medications with a known risk of Torsades de Pointes per crediblemeds.org that cannot be safely discontinued or replaced by a safe alternative.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Baseline	Informed consent, leukapheresis, tumor assessment, organ function testing
Randomization	Day 0	Randomization to Tisagenlecleucel or SOC, start of bridging/salvage therapy
Interim	Week 6	Assessment of disease burden/suitability for CAR-T infusion or autologous HSCT
Primary Assessment	Week 12 (+/- 1 wk)	Primary EFS Assessment via Blinded Independent Review Committee (BIRC)
End of Study	Up to 5 Years	Long-term follow-up for OS, EFS, and patient-reported outcomes

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Event-free survival (EFS) - defined as the time from the date of randomization to the date of the first documented disease progression or stable disease at or after the week 12 (+/- 1 week) assessment, or death due to any cause. **Measurement Method:** Assessed by Blinded Independent Review Committee (BIRC) per Lugano criteria.

7.2 Secondary & Exploratory Endpoints

1. Event Free Survival (EFS) as Assessed by Local Investigator
2. Overall Survival (OS)
3. Overall Response Rate (ORR)
4. Duration of Response (DOR)
5. Patient-Reported Outcomes (Time to definitive deterioration in SF-36v2, FACT-Lym, EQ-VAS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for standard oncology AEs, with strict observation for cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS). **Serious Adverse Events (SAEs):** Safety profile tracking, reporting SAEs within the standard 24-hour timeframe to regulatory bodies. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee handles routine evaluation of patient safety and tolerability signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary effectiveness outcomes (EFS, OS). Safety Set for patients receiving any element of study treatment. **Sample Size**

Justification: Target enrollment of $N=331$ patients powered to detect meaningful statistical difference in event-free survival between CAR-T infusion and standard salvage/HSCT pathways.

Handling of Missing Data: Standard oncology censoring applied for patients without an EFS event at the date of last adequate disease assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Strict on-site and remote monitoring to ensure protocol compliance, especially regarding logistics around autologous T-cell collection, modification, and re-infusion.

Audit Trail: Robust centralized eCRF management validating clinical and BIRC radiographic review checkpoints.

11. Study Identifiers & Governance

Primary ID: CCTL019H2301 **Registry Number:** NCT03570892 **Sponsor/**

Lead Organization: Novartis Pharmaceuticals **Study Title:**

Tisagenlecleucel in Adult Patients With Aggressive B-cell Non-Hodgkin Lymphoma (BELINDA) **Official Regulatory Title:**

Tisagenlecleucel Versus Standard of Care in Adult Patients With Relapsed or Refractory Aggressive B-cell Non-Hodgkin Lymphoma: A Randomized, Open Label, Phase III Trial (BELINDA)

12. Clinical Framework (Lymphoma Specific)

Disease Area / Condition: Non-Hodgkin Lymphoma **Diagnosis**

Threshold: Relapsed or refractory disease within 365 days of first-line immunochemotherapy **Medical Methodology:** Autologous CD19-directed chimeric antigen receptor (CAR) T-cell therapy

Optimization Target: Curative-intent rescue through robust,

durable EFS **Ontology Codes:** MeSH: D008228 | SNOMED: 1172592001

13. Study Procedures & Methodology

Management Pathway: Leukapheresis $\rightarrow$ Bridging chemo $\rightarrow$ Lymphodepletion $\rightarrow$ CAR-T infusion (Experimental) OR Salvage chemo $\rightarrow$ High-dose chemo $\rightarrow$ Autologous HSCT (SOC). **Education Protocol (HCP):** Extensive training on identifying and treating CRS and ICANS.

Education Protocol (Patient): Strict guidelines on recognizing post-infusion toxicities, infection prevention, and ensuring close proximity to the treatment center. **Quality Audit Mechanism:** BIRC radiographic review; stringent chain-of-identity protocols for handling, shipping, and administering living cellular therapy.

14. Observation & Follow-Up Schedule

Total Duration: Up to 5 years follow-up per patient for overall survival and prolonged safety tracking. **Onsite Visit Cadence:** Leukapheresis, bridging therapy, pre-infusion clearance, day of infusion, week 6, week 12, followed by regular interval disease assessments. **Remote Monitoring Frequency:** Continuous evaluation via clinic contact per protocol timeline. **Checklist Review Interval:** Regular PRO assessments (SF-36v2, FACT-Lym) tracking long-term quality of life.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				